

VIVEKANAND S NAIK

Software Engineer | Agentic AI & Backend Systems Developer

+91 8217516956 | Vivekanandnaik333@gmail.com | github.com/Vivekanand-s-naik | linkedin.com/in/Vivekanand-naik

SUMMARY

Computer Science undergraduate specializing in Agentic AI and backend systems engineering. Experienced designing autonomous, tool-using AI agents and multi-agent orchestration pipelines with LangChain and Retrieval-Augmented Generation (RAG), and building production-style REST APIs in Python and Java. Skilled in integrating computer vision and LLM-based reasoning into scalable, full-stack systems, with a strong foundation in data structures, algorithms, databases, and system design.

TECHNICAL SKILLS

Agentic AI & LLM Orchestration: LangChain, Agent Design, Tool/Function Calling, Retrieval-Augmented Generation (RAG), Vector Databases, Multi-Agent Workflows, Prompt Engineering, LLM APIs

Languages: Python, JavaScript, Java (Core & Advanced), SQL, C

Backend & Web: Flask, FastAPI, REST API Design, Spring Boot(basic),HTML5, Tailwindcss, React, CSS3

AI / Machine Learning & Computer Vision: TensorFlow, Keras, PyTorch, OpenCV, MediaPipe, dlib, face_recognition, YOLO

Databases: MySQL, PostgreSQL, MongoDB

Tools & Platforms: Git, GitHub, IntelliJ IDEA, Eclipse, VS Code, Firebase, Postman

Core Concepts: Data Structures & Algorithms, OOP, System Design Fundamentals, REST APIs, Authentication

PROJECTS

ParallaxAI | Agentic AI Assistant & Workflow Automation Platform

Agentic AI • LLM Orchestration • Backend

- Designed and built an autonomous, tool-using AI agent with a modular, backend-first architecture (FastAPI/Flask), enabling the agent to plan tasks, call external APIs, and orchestrate multi-step workflows via LangChain.
- Implemented Retrieval-Augmented Generation (RAG) with a vector database to ground agent responses in external knowledge, improving accuracy and reducing hallucination.
- Engineered authentication and database integration to support scalable, multi-user agentic AI workflows in production-style conditions.

Tech Stack: Python, LangChain, LLM APIs, FastAPI, Flask, Vector Database (RAG), React, REST APIs

Intelligent AI Surveillance & Real-Time Vision Analytics Platform

AI/ML • Computer Vision • Backend

- Built a real-time surveillance platform unifying facial recognition, activity analysis, gaze tracking, and spoof detection into a single multi-model inference pipeline.
- Engineered anti-proxy/liveness detection and I3D-based suspicious-activity detection on live video streams.
- Exposed functionality via REST APIs (Flask/FastAPI) and built a Pandas/Plotly analytics dashboard for real-time monitoring.

Tech Stack: Python, OpenCV, TensorFlow/Keras, PyTorch, MediaPipe, face_recognition, dlib, Flask, FastAPI, Pandas, Plotly

CivicEye | AI-Driven Smart City Infrastructure Monitoring Platform

Computer Vision • Geospatial • Backend

- Built an AI-driven platform that automatically detects civic issues (potholes, garbage, damaged roads, waterlogging) from image/video data using YOLO object detection, integrated with geolocation/GIS mapping.
- Built REST API backend services with a reward-and-penalty rule engine, and reporting/analytics features to help authorities prioritize issues by severity and location.

Tech Stack: Python, OpenCV, TensorFlow/PyTorch, YOLO, FastAPI, Flask, MongoDB/MySQL, React, GIS/Mapping APIs

EDUCATION

Bachelor of Engineering, Computer Science

2023 - 2027

Srinivas University Institute of Engineering and Technology, Mangalore, Karnataka | CGPA: 9.1

Pre-University Course (PCMC)

2023

Kendriya Vidyalaya, Karwar, Karnataka | 70%

CERTIFICATIONS

- IBM Certified - SQL & RDBMS
- IBM Certified - Java Programming
- Ethnotech Certified - Data Science

ACHIEVEMENTS

- Won a competitive off-campus hackathon by designing and building an innovative solution under tight time constraints.
- Organized a college-level hackathon (registrations, evaluation, logistics, judging) and anchored technical/cultural events, demonstrating leadership and public speaking.

ADDITIONAL INFORMATION

Areas of Interest: Agentic AI, AI/ML, Full Stack Development, Data Science | **Interests:** Travelling, Teaching, Cooking